Table 5-7 Critical Values for the χ^2 Distribution

	Probability of greater value P							
ν	.50	.25	.10	.05	.025	.01	.005	.001
1	.455	1.323	2.706	3.841	5.024	6.635	7.879	10.828
2	1.386	2.773	4.605	5.991	7.378	9.210	10.597	13.816
3	2.366	4.108	6.251	7.815	9.348	11.345	12.838	16.266
4	3.357	5.385	7.779	9.488	11.143	13.277	14.860	18.467
5	4.351	6.626	9.236	11.070	12.833	15.086	16.750	20.515
6	5.348	7.841	10.645	12.592	14.449	16.812	18.548	22.458
7	6.346	9.037	12.017	14.067	16.013	18.475	20.278	24.322
8	7.344	10.219	13.362	15.507	17.535	20.090	21.955	26.124
9	8.343	11.389	14.684	16.919	19.023	21.666	23.589	27.877
10	9.342	12.549	15.987	18.307	20.483	23.209	25.188	29.588
11	10.341	13.701	17.275	19.675	21.920	24.725	26.757	31.264
12	11.340	14.845	18.549	21.026	23.337	26.217	28.300	32.909
13	12.340	15.984	19.812	22.362	24.736	27.688	29.819	34.528
14	13.339	17.117	21.064	23.685	26.119	29.141	31.319	36.123
15	14.339	18.245	22.307	24.996	27.488	30.578	32.801	37.697
16	15.338	19.369	23.542	26.296	28.845	32.000	34.267	39.252
17	16.338	20.489	24.769	27.587	30.191	33.409	35.718	40.790
18	17.338	21.605	25.989	28.869	31.526	34.805	37.156	42.312
19	18.338	22.718	27.204	30.144	32.852	36.191	38.582	43.820
20	19.337	23.828	28.412	31.410	34.170	37.566	39.997	45.315
21	20.337	24.935	29.615	32.671	35.479	38.932	41.401	46.797
22	21.337	26.039	30.813	33.924	36.781	40.289	42.796	48.268
23	22.337	27.141	32.007	35.172	38.076	41.638	44.181	49.728
24	23.337	28.241	33.196	36.415	39.364	42.980	45.559	51.179
25	24.337	29.339	34.382	37.652	40.646	44.314	46.928	52.620
26	25.336	30.435	35.563	38.885	41.923	45.642	48.290	54.052
27	26.336	31.528	36.741	40.113	43.195	46.963	49.645	55.476
28	27.336	32.020	37.916	41.337	44.461	48.278	50.993	56.892
29	28.336	33.711	39.087	42.557	45.722	49.588	52.336	58.301
30	29.336	34.800	40.256	43.773	46.979	50.892	53.672	59.703
31	30.336	35.887	41.422	44.985	48.232	52.191	55.003	61.098
32	31.336	36.973	42.585	46.194	49.480	53.486	56.328	62.487
33	32.336	38.058	43.745	47.400	50.725	54.776	57.648	63.870
34	33.336	39.141	44.903	48.602	51.966	56.061	58.964	65.247
35	34.336	40.223	46.059	49.802	53.203	57.342	60.275	66.619
36	35.336	41.304	47.212	50.998	54.437	58.619	61.581	67.985
37	36.336	42.383	48.363	52.192	55.668	59.893	62.883	69.346
38	37.335	43.462	49.513	53.384	56.896	61.162	64.181	70.703
39	38.335	44.539	50.660	54.572	58.120	62.428	65.476	72.055
40	39.335	45.616	51.805	55.758	59.342	63.691	66.766	73.402
41	40.335	46.692	52.949	56.942	60.561	64.950	68.053	74.745
42	41.335	47.766	54.090	58.124	61.777	66.206	69.336	76.084
43	42.335	48.840	55.230	59.304	62.990	67.459	70.616	77.419
44	43.335	49.913	56.369	60.481	64.201	68.710	71.893	78.750
45	44.335	50.985	57.505	61.656	65.410	69.957	73.166	80.077
46	45.335	52.056	58.641	62.830	66.617	71.201	74.437	81.400
47	46.335	53.127	59.774	64.001	67.821	72.443	75.704	82.720
48	47.335	54.196	60.907	65.171	69.023	73.683	76.969	84.037
49	48.335	55.265	62.038	66.339	70.222	74.919	78.231	85.351
50	49.335	56.334	63.167	67.505	71.420	76.154	79.490	86.661

Source: Adapted from J. H. Zar, *Biostatistical Analysis*, 2d ed, Prentice-Hall, Englewood Cliffs, N.J., 1984, pp. 479–482, table B.1. Used by permission.

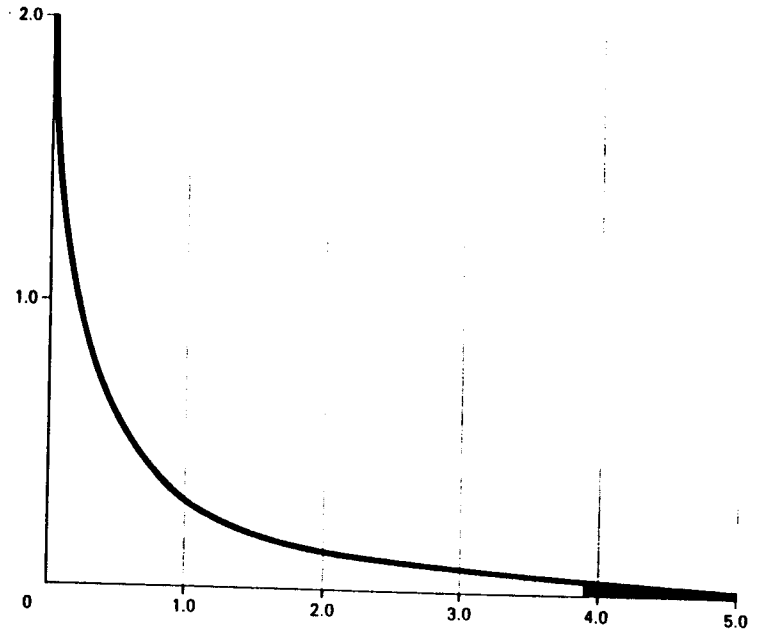


Figure 5-7 The chi-square distribution with 1 degree of freedom. The shaded area denotes the biggest 5 percent of possible values of the χ^2 test statistic when there is no relationship between the treatments and observations.